Orthographic Gaussian Splatting for Volumetric Anatomical Visualization

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Research Objectives & Questions

Research Objective: Investigate the feasibility of using Gaussian splatting for 3D visualization of medical imagery (CT scans).

Research Questions:

- How can Gaussian splatting be adapted for medical imaging?
- What challenges arise in terms of camera calibration and image alignment?
- Can Gaussian splatting improve the accuracy of 3D reconstructions compared to traditional methods?

Introduction to Gaussian Splatting

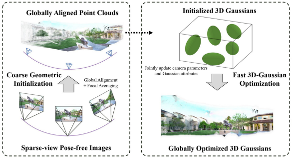
What is Gaussian Splatting?

A technique for rendering 3D data by representing points in space as Gaussians.

Commonly used in computer graphics for realistic visualization and modeling.

Why it's important for medical imaging?

Could offer a dynamic, interactive way to visualize complex medical datasets, such as CT scans.



Primary Elements of a Gaussian Splat

A Gaussian splat consists of the following primary elements:

Mean (Center): The central location of the Gaussian function in the 3D space, typically defined as the position of the splat.

Covariance Matrix (Shape): Describes the spread and orientation of the Gaussian distribution. It defines how the splat will deform and influence surrounding areas.

Weight (Intensity): The magnitude or intensity of the splat, which determines its effect on the surrounding environment. A higher weight results in a stronger influence.

Radius (Spread): The spatial extent of the splat, typically defined by the standard deviation of the Gaussian. A larger radius means the splat has a broader influence, while a smaller radius creates a more focused splat.

Research Motivation

Current Challenges in Medical Imaging:

3D visualization methods for medical data are limited.
Traditional rasterization techniques often fail to provide sufficient detail or realism.

Goal:

To explore if Gaussian splatting can address these limitations by offering a more realistic and flexible way to visualize 3D medical data.

1. Cinematic Gaussian Splatting (Cinematic GS)

Cinematic GS compresses large CT/MRI anatomy scans into lightweight 3D Gaussian representations, enabling real-time, photorealistic rendering. High-resolution scans are rendered offline using path tracing, then optimized using techniques like Mip-splatting for level-of-detail optimization. These compact models, typically under 70 MB even for multi-gigabyte datasets, maintain fine details comparable to the original voxel resolution and can achieve 60 FPS rendering on mobile or VR hardware. The pipeline involves selecting camera poses for anatomical structure capture, differentiable rendering training and model compression (e.g., via vector quantization).

Strengths: High fidelity, compact memory footprint.
Limitations: Baked rendering parameters (transfer functions, lighting), requiring separate models for changes.

2. GaSpCT (Gaussian Splatting for CT)

GaSpCT applies 3D Gaussian Splatting to sparse-view CT reconstruction. Key techniques:

- Specialized loss functions like beta sparsity and total variation loss for foreground/background separation.
- Uniform Gaussian initialization based on the expected anatomy.

GaSpCT avoids traditional Structure-from-Motion (SfM) approaches, allowing for interpolation between views, reducing scan time, and minimizing radiation exposure without sacrificing image quality.

Strengths: Efficient training and reduced scan burden.
Limitations: Primarily validated on simulated data, with real-world challenges (e.g., patient motion, curved detectors) not fully addressed.

3. Multi-Layer Gaussian Splatting

Multi-Layer GS builds on the cinematic approach, enabling immersive VR exploration of anatomy using layered 3D Gaussian models. The method trains Gaussian splats in stages:

Render the scan using one transfer function (e.g., bones) and train the first layer.

Render with a second transfer function (e.g., muscles) and train the second layer.

Prune inactive Gaussians and cluster similar ones across layers for compression.

During VR rendering, users can toggle, clip, or combine layers without retraining, enhancing the interactive experience.

Strengths: Interactive frame rates on VR hardware with good anatomical detail.
Limitations: Static model structure, potential artifacts when overlapping layers or zooming into low-contrast regions.

Comparative Summary of 3D Gaussian Splatting Methods

Aspect	Cinematic GS	GaSpCT	Multi-Layer GS
Purpose	Real-time photo-realistic rendering	Sparse-view CT reconstruction	Immersive VR exploration
Input Data	High-resolution CT/MRI scans	Sparse X-ray projections	CT/MRI scans
Model Size	<70 MB	Variable, depends on dataset	Varies based on layers
Strengths	High fidelity, low memory footprint	Reduced scan time, low radiation exposure	Interactive VR, good anatomical detail
Limitations	Fixed rendering parameters	Validated on simulated data only	Static model structure, possible artifacts
Rendering Performance	60 FPS on mobile/VR hardware	High quality, depends on reconstruction	Interactive frame rates
Interactivity	Low	Low	High, toggle layers in VR
Compression	Mip-splatting, vector quantization	None (sparse data input)	Layer-based pruning and clustering

Table: Comparison of 3D Gaussian Splatting Methods in Medical Imaging

Comparison with Base Gaussian Splatting Pipeline (Part 1)

Aspect	Base Gaussian Splatting	Cinematic GS	GaSpCT	Multi-Layer GS
Purpose	General-purpose Gaussian splatting for 3D data visualization	Real-time photo-realistic rendering	Sparse-view CT reconstruction	Immersive VR exploration
Input Data	Full 3D volumetric data (e.g., voxel grids)	High-resolution CT/MRI scans	Sparse X-ray projections	CT/MRI scans
Use Case	Visualization of 3D volumes (medical, geological, etc.)	Medical rendering, photorealism for VR	Medical CT reconstruction with reduced scan time	VR anatomical exploration
Strengths	General-purpose flexibility	High fidelity, compact representation, mobile/VR rendering	Reduced scan time, low radiation exposure	Interactive VR, multiple anatomical layers
Limitations	High memory usage, slower rendering for large volumes	Fixed rendering parameters	Validated on simulated data only	Static model, possible artifacts with zoom

Table: Comparison of 3D with Base Gaussian Splatting Pipeline (Part 1)

Comparison with Base Gaussian Splatting Pipeline (Part 2)

Aspect	**Base Gaussian Splatting**	**Cinematic GS**	**GaSpCT**	**Multi-Layer GS**
Performance	Moderate rendering speed, depends on dataset size	60 FPS on mobile/VR hardware	High quality, depends on sparse data	Interactive frame rates in VR
Interactivity	Low interactivity, static models	Low interactivity	Low interactivity	High interactivity, layer toggling
Compression	None, standard Gaussian fitting	Mip-splatting, vector quantization	None (sparse data input)	Layer-based pruning and clustering

Table: Comparison of 3D Gaussian Splatting Methods with Base Gaussian Splatting Pipeline (Part 2)

Methodology

Dataset Selection:

Raw CT slices (hip-CT, 1800 images) chosen after evaluating available datasets.

Converted from JPEG2000 to PNG via TIFF format.

Codebase Adaptation:

Adapted `Gaussian Splatting` pipeline to work with medical data.

Resolved dependency conflicts with CUDA, PyTorch, and Visual Studio.

Structure-from-Motion (SfM):

Used M60 tank dataset to generate camera extrinsics due to challenges with CT slices.

SfM couldn't correlate CT images; switched to a non-medical dataset to test pipeline.

Gaussian Splatting:

Initial point cloud generated, but results were blurry.

Adjusted camera extrinsics for orthographic projection, leading to more structured output.

Pipeline Setup

Pipeline Source:

Retrieved codebase from *Gaussian Splatting for Cinematic Anatomy*.

Dataset:

Used 1800 slices of CT scan images (hip-CT), converted to PNG format.

Code Adaptations:

Addressed missing dependencies and removed deprecated code.

Testing:

Successfully ran Gaussian splatting on M60 tank dataset to validate pipeline.

Dataset Selection & Preprocessing

Dataset Selection:

Initially explored X-ray and CT datasets but faced compatibility issues.
Chose *hip-CT*, a dataset with 1800 slices of brain imagery (partial side).

Preprocessing:

Converted raw CT scan images from JPEG2000 (*JP2*) format to *PNG* (8-bit).
Used JPEG2000 and TIF formats for intermediate conversion.

Data Compatibility:

Adapted dataset for Gaussian splatting pipeline by resolving file format issues.

Structure-from-Motion (SfM) Challenges

SfM Requirements:

SfM typically requires common features (edges or textures) across images for camera extrinsics calculation.

Essential for generating 3D structures, such as point clouds, from 2D slices.

Challenges with Medical Datasets:

CT Slices: Medical CT scan slices lack common features between adjacent slices (e.g., no shared edges or textures).

Difficulty in aligning and correlating images for SfM due to the uniform nature of medical imaging.

SfM Failure on hip-CT Dataset:

Attempted SfM on 1800 slices from *hip-CT* dataset resulted in only 2 images being correlated.

The absence of detectable features across slices prevents effective 3D reconstruction.

Alternative Approaches:

Explored using non-medical datasets (e.g., M60 tank)

Exploring Alternative Datasets

Challenges with Medical Datasets:

Medical datasets (e.g., *hip-CT*) lacked common features required for Structure-from-Motion (SfM).

The uniformity and lack of textures across CT slices hindered image alignment and feature extraction.

Switch to Non-Medical Datasets:

M60 Tank Dataset: Selected as an alternative to test the pipeline due to its rich features and available SfM data.

300 images of the M60 tank provided sufficient common features for successful SfM processing using COLMAP.

SfM Success with M60 Tank Dataset:

SfM process successfully generated both camera intrinsics and extrinsics, solving the camera calibration problem.

The dataset allowed for accurate 3D point cloud generation, providing proof of concept for the Gaussian splatting pipeline.

Results from M60 Tank Dataset

figure: Initial random point cloud generated from M60 Tank dataset.

Key Steps:

- **SfM Process:** COLMAP successfully solved camera parameters (intrinsics extrinsics) for 300 images.
- **Initial Point Cloud:** 22.33 minutes of training resulted in a random point cloud.
- **Visualization:** Web-based tools used to visualize the Gaussian splat, showing blurry results.
- **Adaptation:** Focus shifted to adapting for the medical dataset with custom camera extrinsics.

Adapting to Medical Data

Initial Struggles

Dataset Selection: Struggled to find a suitable dataset due to the lack of "sliced" data for our needs.

Raw CT Data: Ultimately worked with raw CT slice data, manually prepared.

Paper Selection Evaluation

Narrowed down to three key papers:

- *GaSpCT*: Gaussian splatting for CT projection synthesis.
- *Multi-Layer Gaussian Splatting*: Immersive anatomy visualization.
- *Gaussian Splatting for Cinematic Anatomy*: Chose the pipeline despite some gaps.

Each paper thoroughly analyzed by team members.

Adapting to Medical Data - 2

Pipeline Setup Trials

Codebase Issues: Faced dependency and environment issues; manual fixes required.

Dataset Conversion: Converted original CT dataset (hip-CT) from JPEG2000 to PNG format.

Pipeline Adjustment: Implemented custom fixes for background images and code errors.

Challenges with SfM

SfM Challenges: Struggled with SfM due to lack of common features between slices.

Alternative Dataset: Used a tank dataset (M60, 300 images) to successfully generate camera data.

Initial Point Cloud: Generated Gaussian splats with random point cloud but blurry results.

Key Findings

Orthographic Projection

- CT scans use orthographic projection, not perspective.
- Required a rewrite of the camera extrinsics script.

SfM Challenges

- SfM struggles due to the lack of common features between slices.
- Utilized an M60 tank dataset to generate camera extrinsics.

Gaussian Splatting Progress

- Initial random point clouds generated via Gaussian splatting.
- Tweaked parameters (iterations, density) for better results.

Implications of the Findings

Impact on Medical Imaging

Orthographic projection better reflects CT scan realities.

adaptation of pipelines to support medical imaging standards.

Challenges for Reconstruction

SfM dependency on common features limits the feasibility for CT slice datasets.

improving handling of images with minimal features.

Gaussian Splatting Potential

Gaussian splatting holds potential for clearer 3D visualization of CT scans.

Refinements in parameters can lead to better medical imaging techniques.

Acknowledging Limitations

Dataset Issues

Limited datasets, resorted to raw CT slices.

Conversion issues (JPEG2000 $\rightarrow$ .png).

Codebase SfM Problems

Incompatible dependencies.

SfM failed due to lack of common features in CT slices.

Pipeline Testing

Gaussian splatting gave random results.

Camera extrinsics (orthographic projection) were manually adjusted.

Suggestions for Future Research

Improving Data Compatibility

Explore more medical datasets, especially DICOM.

Investigate alternative image formats and preprocessing methods.

Enhancing Pipeline Flexibility

Adapt pipeline for a wider range of imaging modalities.

Implement automated SfM solutions for medical datasets.

Optimization of Gaussian Splatting

Explore multi-resolution approaches to improve image clarity.

Fine-tune parameters for better convergence and detail.

Utilizing GPU-based Tools

Develop optimized rasterization methods for medical datasets.

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